MARAM JEBALI

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Profile

AI & Data Science Master's student (ESPRIT, EUR-ACE, expected 2027) with **3 AI internships** building ML, computer-vision and multimodal LLM systems on geospatial, environmental and health data. Delivered **end-to-end pipelines** combining knowledge graphs, RAG and multi-agent architectures, including a **6-server MCP system** for chemical risk analysis. Seeking to contribute a rigorous, data-driven perspective to sustainability, environmental monitoring and resilient infrastructure.

Experience

AI Intern – Multimodal Internal AI Assistant & Document Processing Aug. – Sep. 2026
Orange *Tunisia*
- Developed **Oli**, an internal multimodal RAG assistant, owning the end-to-end retrieval pipeline from document ingestion to grounded generation.

Stack: Next.js, React, TypeScript, Vercel AI SDK, Qdrant, RAG, VLMs, OpenCV, Gemini, Groq, Mistral, PostgreSQL

Generative AI Intern – AI-Assisted Urban Planning Platform Jul. – Aug. 2026
Talan SummerCamp *Tunisia*
- Co-developed and deployed **UrbanMAIND**, an AI-assisted urban-planning platform using **geospatial data** and French regulations for feasibility studies, macro-zoning and spatial reasoning.
- Implemented **LangGraph multi-agent workflows** for site analysis, **regulatory RAG** and 2D mass planning, translating complex geospatial and regulatory inputs into structured planning outputs.

Stack: React, FastAPI, Mistral, LangGraph, ChromaDB, RAG, AI Agents, Geospatial APIs, PLU, GeoJSON, Vercel, Render

AI Intern – Autonomous Evaluation of Generated Content & Image Relevance Jun. – Jul. 2025
Value *Tunisia*
- Built a **multimodal evaluation workflow** (LLM-as-a-judge, RAGAS) to assess generated tourism content and image relevance for downstream pipelines.

Stack: Python, LLM-as-a-Judge, RAGAS, NLP, Multimodal Evaluation, LangChain, NumPy, Matplotlib

Selected Projects & Hackathons

OVERDOSE – AI-Based Chemical Exposure & Risk Analysis Feb. 2026 – Present
Link to Problem Presentation
- Co-developed an **AI risk-assessment system** analysing chemical exposure in everyday consumer products, converting ingredient data into actionable safety insights.
- Designed the **Biological Intelligence Layer**: an MCP agent orchestrating **6 specialised servers** over a **Neo4j Aura knowledge graph**, adding combination, cumulative-exposure, risk-ranking and synergistic-effect analysis.

Stack: Python, MCP Server, Neo4j Aura, Knowledge Graphs, LLMs, Semantic Retrieval, Sentence Transformers

CancerCare+ – AI-Powered Breast Cancer Support Platform Jul. – Oct. 2025
IEEE Women in Engineering (WIE) · Pink October Initiative
- Developed **YOLOv8/CNN** breast-cancer detection on mammography with **Grad-CAM** transparency; contributed to a patient–doctor support platform through an **IEEE WIE** community initiative.

Stack: YOLOv8, CNN, Grad-CAM/XAI, CBIS-DDSM

SAHHTAEK – Intelligent Medical Guidance & Health Decision Support Sep. 2025 – Feb. 2026
Academic Project
- Designed the **multi-agent orchestration (LangGraph)** and built a **RAG assistant** for medical misinformation correction with NLP-based symptom analysis.

Stack: Python, Django, LangChain, LangGraph, RAG, NLP, LLMs, Multi-Agent Systems

Certifications

NVIDIA DLI (5): Fundamentals of Deep Learning (2024) · Generative AI with Diffusion Models (2025) · Multimodal AI Agents (2025) · Anomaly Detection (2026) · Predictive Maintenance (2026)
AWS: AWS Academy Cloud Foundations (2026)

Leadership & Activities

Workshop Instructor & Technical Mentor 2024 – Present
DeepFlow AI Club | IEEE CIS · WIE *Tunisia*
- Design and deliver **hands-on AI/ML workshops**, translating complex topics into project-oriented sessions.
- Mentor students and teams on **AI/ML project design**; collaborate with **IEEE CIS and WIE volunteers** on community knowledge-sharing.

Education

Master's degree in Computer Science – AI Specialisation 2022 – 2027
ESPRIT School of Engineering (EUR-ACE label) *Tunisia*

Technical Skills

AI/ML: Machine Learning, Deep Learning, PyTorch, TensorFlow, Scikit-learn, NLP, Computer Vision, Statistical Analysis, R
GenAI & LLM: LLMs, RAG, LangChain, LangGraph, Multi-Agent Systems, MCP, Transformers, Semantic Search
Programming: Python, SQL, Java, JavaScript, C, C++
Data & Analytics: Pandas, NumPy, Power BI, SSIS/SSMS, Hadoop, Geospatial Analysis (GeoJSON, spatial reasoning)
Databases & Infrastructure: Qdrant, Neo4j, PostgreSQL, FAISS, Git/GitHub, AWS, MLOps

Languages

Arabic (native) · French (C1) · English (C1) · German (A2)